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Activation Matching for Minimalist Explanation Generation

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 21 Nov 2025 (modified: 25 Nov 2025)  AAAI 2026 Workshop EGSAI Submission  EGSAI, Reviewers, Authors
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Keywords: Activations, Explanations, Interpretability, Minimality

TL;DR: This paper presents an activation matching based approach for minimalist explanation generation.

Abstract:

In this paper we introduce an activation-matching--based approach to generate minimal, faithful explanations for the decision-making of a pretrained classifier on any given image. Given an input image (x) and a frozen model (f), we train a lightweight autoencoder to output a binary mask (m) such that the explanation ($e = m \odot x$) preserves both the model's prediction and the intermediate activations of (x). Our objective combines: (i) multi-layer activation matching with KL divergence to align distributions and cross-entropy to retain the top-1 label for both the image and the explanation; (ii) mask priors---L1 area for minimality, a binarization penalty for crisp 0/1 masks, and total variation for compactness; and (iii) abductive constraints for faithfulness and necessity. Together, these objectives yield small, human-interpretable masks that retain classifier behavior while discarding irrelevant input regions, providing practical and faithful minimalist explanations for the decision making of the underlying model.

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Paper Decision

Decision by Program Chairs 📅 06 Dec 2025, 14:23 (modified: 06 Dec 2025, 14:38) 👁 Program Chairs, Authors

📄 Revisions (/revisions?id=dwVsPPFPa6)

Decision: Accept - Conditional

Comment:

Dear Author,

We are pleased to inform you that your submission has been (conditionally) accepted to the AAAI26 EGSAI - Community Activity Workshop. Congratulations! We received an overwhelmingly positive response for our inaugural edition, with 107 submissions, and yours is among 64 selected. Reviewer comments are available on OpenReview.

Camera-Ready Abstract (Required) Your revised abstract may include: Up to 2 pages of main content, including brief sections for Introduction, Method/Approach, Experiments and Results, and Conclusion. Up to 2 additional pages for references, acknowledgments, and appendix (additional information and results).

Please ensure that: All co-authors are listed (not only the student). The camera-ready PDF is not anonymized (AAAI guidelines). The PDF includes a link to your pre-recorded YouTube video for 3MT finalist shortlisting.

Mandatory Pre-Recorded Video Submission To change your status from Accept-Conditional → Accept, you must submit a pre-recorded presentation video.

Video Requirements: Maximum duration: 3 minutes (per 3MT rules: <https://threeminutethesis.uq.edu.au/resources/3mt-competitor-guide> (<https://threeminutethesis.uq.edu.au/resources/3mt-competitor-guide>)) Format: MP4 Must include one static slide only (per 3MT rules). No animations, transitions, or props Only the speaker video may contain motion. Speaker video may be embedded, preferably at the bottom-right May use a virtual laser pointer/mouse to highlight key information Recording Tools: PowerPoint recorder is recommended for simplicity. You may use any tool of your choice as long as the rules above are followed. File Naming Format: AAAI26_EGSAI_[object Object] (Example: AAAI26_EGSAI_123.mp4) Submissions that violate 3MT rules will be disqualified.

All camera-ready files and video submissions must be uploaded via the Google Form by: December 20, 2025 (AoE). A separate email with the Google Form link will be sent shortly.

We look forward to your participation and to showcasing your research at the AAAI26 - EGSAI Community Activity Workshop.

Best regards, AAAI26 - EGSAI Program Chairs



Solid Contribution to Minimalist XAI with Room for Methodological Rigor and Quantitative Validation

Official Review by Reviewer xscW 📅 29 Nov 2025, 12:16 (modified: 06 Dec 2025, 16:04) 👁 Program Chairs, Reviewer xscW, Authors

📄 Revisions (/revisions?id=sFIjII7jyD)

Review:

Summary

This paper presents an activation-matching approach for generating minimal, faithful explanations of image classifier decisions through learned binary masks. The core methodology—training a lightweight autoencoder to preserve both predictions and intermediate activations while enforcing sparsity—is conceptually sound and addresses a genuine need in explainable AI. The visual results are compelling, demonstrating the method's ability to produce highly

sparse explanations (2-5% pixel retention) that maintain classification confidence. However, the paper suffers from insufficient quantitative evaluation, limited experimental scope, and missing technical details that prevent full assessment of its claims and reproducibility.

Major Issues

1. Absence of Quantitative Evaluation Metrics

The paper relies almost exclusively on visual demonstration without systematic quantitative analysis:

- **No fidelity metrics:** What percentage of explanations preserve the original top-1 prediction? What is the distribution of confidence changes?
- **No minimality benchmarks:** How does the achieved sparsity (% active pixels) compare across images, classes, or datasets? Are 2-5% typical or cherry-picked examples?
- **No comparison with baselines:** LIME, Grad-CAM, SHAP, and other attribution methods are cited but never compared. How does activation matching perform against these established techniques in terms of sparsity, fidelity, or human interpretability?
- **No robustness analysis:** The "abductive constraint" enforces robustness to background perturbations during training, but no evaluation demonstrates whether this actually improves explanation stability compared to methods without this constraint.

Recommendation: Include quantitative tables reporting mean/std metrics across a test set (e.g., 1000 ImageNet validation images): fidelity rate, sparsity (% pixels), confidence preservation, and comparison with at least 2-3 baseline methods.

2. Insufficient Experimental Scope

- **Single dataset:** Only ImageNet is evaluated. Medical imaging, natural language (via visual transformers), or other safety-critical domains would strengthen generalization claims.
- **Limited architecture diversity:** While ResNet-18, ViT, and MobileNet are mentioned, actual results appear only for ResNet-18. Do the explanations differ meaningfully across architectures? Does activation matching work equally well for CNNs vs. transformers?
- **No failure case analysis:** What happens when the method fails? Are there classes or image types where explanations are uninformative or the mask converges to trivial solutions?

Recommendation: Expand evaluation to at least one additional domain (e.g., medical imaging) and provide systematic results across all three mentioned architectures.

3. Missing or Underspecified Technical Details

Several critical methodological components lack sufficient detail for reproducibility:

- **Autoencoder architecture:** "Simple U-Net-based" is vague. How many layers? What skip connections? What activation functions for the mask output (sigmoid $\rightarrow$ binarization)?
- **Loss function weights:** The composite loss combines 6+ terms (activation matching across layers, KL divergence, cross-entropy, L1, binarization, TV, robustness). What are the actual coefficients α_l and relative weights? How sensitive is performance to these hyperparameters?
- **Training details:** Learning rate, optimizer, number of iterations, convergence criteria—none are specified.
- **Binarization mechanism:** $L_{bin} = \|m - m^2\|_1$ encourages binary values during training, but are masks actually binarized at inference (e.g., thresholding)? If so, at what threshold?
- **Perturbation generation:** For the robustness constraint, what distribution is r sampled from? Gaussian noise? Random ImageNet patches?

Recommendation: Add a technical appendix or supplementary material with complete architecture diagrams, loss weight schedules, and training hyperparameters.

4. Insufficient Validation of "Faithfulness" Claims

The paper claims explanations are "faithful" because they preserve activations and predictions, but this is not rigorously validated:

- **Sufficiency vs. necessity:** Does the method verify that removed pixels ($1 - m$) are truly irrelevant? The robustness term addresses this partially, but no explicit necessity test is shown.
- **Activation matching as proxy:** Why assume intermediate activation similarity guarantees faithful explanations? A discussion of this assumption's limitations is needed.
- **Shortcut detection claim:** Figure 2 shows the model highlighting a leash, interpreted as "shortcut learning." While plausible, this interpretation lacks validation—was this behavior confirmed through ablation (e.g., removing leash-containing images from training)?

Recommendation: Include ablation studies demonstrating that predictions fail when masked regions are perturbed, and that unmasked regions can be arbitrarily modified without prediction change.

Minor Issues

5. Writing and Presentation

- **Abstract ambiguity:** "abductive constraints for faithfulness and necessity" is mentioned but not clearly defined in the abstract. The term "abductive" typically refers to reasoning from observations to hypotheses—how does this apply to the robustness term?
- **Figure quality:** Figure 2's image quality is poor (compressed/blurry), making it difficult to assess the explanations. High-resolution supplementary figures would help.
- **Notation inconsistency:** Page 2 uses $e^\sim$ for perturbed explanations but does not clearly distinguish it from e initially.
- **Related work integration:** The introduction surveys XAI literature but does not identify specific gaps this work fills. How does activation matching improve upon or differ from existing minimal explanation methods?

6. Limited Discussion of Limitations

- **Computational cost:** Training an autoencoder per image (or per class?) could be expensive. What is the inference-time cost compared to gradient-based methods like Grad-CAM?
- **Hyperparameter sensitivity:** Figure 2 demonstrates that varying loss weights produces dramatically different explanations. How should practitioners choose these weights? Is there a principled approach or is it ad-hoc tuning?
- **Multi-object handling:** The multi-object example (Figure 1) is interesting but underdeveloped. Does the method automatically produce separate masks for each instance, or does it require manual intervention?

7. Future Work Needs Specificity

The conclusion mentions "formal guarantees of minimality" and "identifying sparse sub-circuits" as future directions, but these are extremely vague. What specific theoretical frameworks or experimental approaches are envisioned?

Questions for Authors

1. **What is the fidelity rate across a representative test set?** Please report the percentage of explanations that preserve the original top-1 prediction.

- 2. **How does performance compare to LIME, Grad-CAM, or other baselines?** A quantitative comparison table is essential.
- 3. **What are the exact loss weights and training hyperparameters?** These are critical for reproducibility.
- 4. **How does the method scale?** Is the autoencoder trained once per image, once per class, or amortized across a dataset?
- 5. **Can you provide quantitative evidence for the "shortcut learning" claim in Figure 2?** For example, does removing leash pixels degrade performance on dog classification?
- 6. **What happens for classes with distributed features** (e.g., textures, patterns) rather than localized objects?

Strengths (To Be Fair)

Despite the above concerns, the paper has notable strengths:

- **Novel approach:** Combining activation matching with mask learning is a creative synthesis of ideas from representation learning and XAI.
- **Compelling visualizations:** The other examples effectively demonstrate extreme sparsity with maintained fidelity.
- **Practical relevance:** Minimal explanations have clear value for interpretability and debugging.
- **Flexibility:** The framework appears adaptable across architectures and potentially other modalities.

Verdict and Recommendations

This paper presents a promising approach that makes a solid conceptual contribution to minimalist explainable AI. However, it currently reads more as a technical report or proof-of-concept than a complete research paper. To meet publication standards:

Required for acceptance:

- 1. Comprehensive quantitative evaluation across a test set (fidelity, sparsity, robustness)
- 2. Direct comparison with at least 2-3 baseline XAI methods
- 3. Complete methodological details for reproducibility
- 4. Results across all three mentioned architectures (ResNet, ViT, MobileNet)

Strongly recommended: 5. Evaluation on a second domain beyond ImageNet 6. Ablation studies validating the necessity of each loss component 7. User study or formal faithfulness evaluation 8. Discussion of computational costs and practical deployment

With these additions, this could become a strong contribution to the XAI literature. In its current form, the work is promising but incomplete.

Rating: 6: Marginally above acceptance threshold

Confidence: 4: The reviewer is confident but not absolutely certain that the evaluation is correct



Review

Official Review by Reviewer d7vL 📅 25 Nov 2025, 15:44 (modified: 06 Dec 2025, 16:04) 👁 Program Chairs, Reviewer d7vL, Authors

📄 Revisions (/revisions?id=43cuEguqgb)

Review:

This paper introduces an activation-matching-based approach to generate minimal, faithful explanations for the decision-making of a pretrained classifier on any given image. For that, the approach minimises a multi-layer activation distance, masks are made compact and interpretable, and the system is made robust against perturbations.

The explanation depicted in Figure 1 is noisy and rarely represents the object part. A similar observation has been made for the images presented in Figure 2. Moreover, there is no quantitative analysis. Only 2 visual examples are presented without explanation and comparison with existing benchmark approaches, including prototype-based, LIME, and Grad-CAM.

It is not clear why the explanation must be minimal, although it should be as comprehensive as possible until it accurately represents the genuine reason for a possible classification/misclassification.

Rating: 4: Ok but not good enough - rejection

Confidence: 5: The reviewer is absolutely certain that the evaluation is correct and very familiar with the relevant literature

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